Advanced Encryption Standard 加密
(AES)

AES-128
AES-192
AES-256

密钥扩展.

plaintext [每个小格是1-byte=8bit]

01	23	45	67
89	ab	cd	ef
fe	dc	ba	98
76	54	32	10

key [16 words]

i=0	i=1	i=2	i=3
0f	15	71	c9
47	d9	e8	59
0c	b7	ad	d6
af	7f	67	98

i%Nk==0 特殊处理

temp

12	13	14	15

rot_word

13	14	15	12

S-tab sub_word

13'	14'	15'	12'

coef_add

Rcj	00	00	00

13''	14''	15''	12''

State_1

key_1
W[0:3]

0	4	8	
1	5	9	
2	6	10	
3	7	11	

temp

12	13	14	15

add_round_key

sub_bytes
字节替代

1	2	3	4
1	2	3	4
1	2	3	4
1	2	3	4

查S表=16进制表示时
如: {xy}
x: row
y: column

key_2
W[4:7]

16	20	24	28
17	21	25	29
18	22	26	30
19	23	27	31

shift_rows
行移位

1	2	3	4
2	3	4	1
3	4	1	2
4	1	2	3

每row 1, 2, 3, 4 自己变化

$a(x)=\{02\}+\{01\}x+\{01\}x^2+\{03\}x^3$

mix_columns
列混淆

2	3	1	1
1	2	3	1
1	1	2	3
3	1	1	2

X =

add_round_key

State_2

round 1

=

State_10

sub_bytes

shift_rows

add_round_key

round 10

key_11
W[40.43]

ciphertext

Advanced Encryption Standard 高级加密标准 (AES)

等价的逆向算法 Exchange

① inverse_shiftrows [inverse_sub bytes] = inverse_sub bytes [inverse_shiftrows]

② 逆向列混淆 (Si ⊕ Wi) = [逆向列混淆 (Si)] ⊕ [逆向列混淆 (Wi)]

等价逆密码



